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**EX-802 (GS)**  
**B.E. VIII Semester** Examination, June 2020  
**Grading System (GS)**  
**Computer Applications to Power System**  
*Time : Three Hours*  
*Maximum Marks :70*

**Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) Explain line loadability in brief.  
b) What do you understand by capability curves of alternator?
2. Explain why it is necessary to control of load bus voltage using reactive power control.
3. Explain SVC and SVS in detail.
4. a) Define power system security? Explain security level with the help of flow chart.  
b) Specify the following:
  - i) Contingency analysis
  - ii) Contingency ranking.
5. Differentiate the following:
  - i) Security
  - ii) Stability
  - iii) Reliability
6. a) Explain voltage stability. How is different than angle stability?  
b) Explain V-Q sensitivity.
7. Enlist the various method for voltage stability enhancement and explain any one.

OR

Develop any proximity index for voltage stability assessment.

8. Write short notes on any two of the following:
  - a) Effect of load models on voltage stability of power system
  - b) PV-curve for voltage stability assessment
  - c) Corrective rescheduling for power system security
  - d) Transmission line models

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